

WHERE

vs.

HAVING

why and how to
filter data in SQL

WHERE: what does it do?

Filters individual *rows* based
on specific conditions

Uses comparison operators
like =, >, <, LIKE

WHERE:

how do you use it?

```
SELECT  
    customer_id  
    , order_id  
    , amount  
FROM orders  
WHERE amount > 50;
```



filters out all rows
where the amount is
less than or equal to
\$50

WHERE:

what do results look like?

customer_id	order_id	amount
57489	2662	59.99
57498	2663	59.99
57698	2682	59.99
38696	2683	59.99
57731	2688	59.99
57848	2699	59.99
48093	2701	59.99
57901	2705	59.99
51657	2710	59.99
57961	2714	59.99

each customer_id and order_id pairing on its own row with no amount less than 50

HAVING:

what does it do?

Filters *groups* based on the results of aggregate functions

Uses comparison operators like =, >, <

Must include a GROUP BY

HAVING:

how do you use it?

```
SELECT
  customer_id
  , COUNT(DISTINCT order_id) AS total_orders
  , SUM(amount) AS total_payment
FROM orders
GROUP BY customer_id
HAVING total_payment > 100;
```

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requires a GROUP BY
clause to create groups

filters out customers
who have spent less
than \$100 in orders

HAVING:

what do results look like?

customer_id	total_orders	total_payment
24159	3	149.97
51657	2	109.98
53233	2	109.98
58443	2	109.98
39643	2	109.98
57464	2	109.98
62780	2	109.98
69639	3	149.97
79647	2	109.98
74523	2	109.98

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results grouped by customer_id and
return the total number of orders
and amount spent by that customer

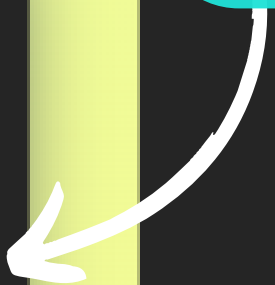
KEY DIFFERENCE: what it filters on

WHERE filters individual rows

HAVING filters groups after aggregation

```
SELECT
  customer_id
  , order_id
  , amount
FROM orders
WHERE amount > 50;
```

amount is an individual value



```
SELECT
  customer_id
  , COUNT(DISTINCT order_id) AS total_orders
  , SUM(amount) AS total_payment
FROM orders
GROUP BY customer_id
HAVING total_payment > 100;
```

total_payment
aggregates the
amount column



KEY DIFFERENCE: what it needs

WHERE doesn't need an additional clause
HAVING requires a GROUP BY clause

```
SELECT
  customer_id
  , order_id
  , amount
FROM orders
WHERE amount > 50;
```

WHERE stands alone

GROUP BY
customer_id

```
SELECT
  customer_id
  , COUNT(DISTINCT order_id) AS total_orders
  , SUM(amount) AS total_payment
FROM orders
GROUP BY customer_id
HAVING total_payment > 100;
```

KEY DIFFERENCE: what it needs, cont.

WHERE can't use aggregate functions
HAVING needs aggregate functions

```
SELECT
  customer_id
  , order_id
  , amount
FROM orders
WHERE amount > 50;
```

WHERE doesn't
use aggregates

HAVING uses
aggregates LIKE
SUM or COUNT

```
SELECT
  customer_id
  , COUNT(DISTINCT order_id) AS total_orders
  , SUM(amount) AS total_payment
FROM orders
GROUP BY customer_id
HAVING total_payment > 100;
```

KEY DIFFERENCE: how it can be referenced

WHERE can't reference an alias

HAVING can reference an alias

```
SELECT
  customer_id
  , order_id
  , amount
FROM orders
WHERE amount > 50;
```

use the column
name in WHERE

use either the
alias or the
aggregate, like
SUM(amount)

```
SELECT
  customer_id
  , COUNT(DISTINCT order_id) AS total_orders
  , SUM(amount) AS total_payment
FROM orders
GROUP BY customer_id
HAVING total_payment > 100;
```